Basics 3: Vector Calculus for Plasma Physics

I - Mathematical and computational foundations

Gradient, divergence, curl, Laplacian, integral theorems, and physical interpretation

Textbook draft, version 1.0. Vector calculus turns spatial variation into physical statements: gradients drive forces, divergence measures sources and compression, and curl measures circulation and current. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

Contents

1 Learning objectives

After completing this note, the reader should be able to:

- Compute and interpret gradient, divergence, curl, and Laplacian operators.
- Derive local conservation laws from integral balances.
- Use the divergence theorem and Stokes theorem.
- Apply vector identities used in MHD force balance and induction.
- Recognize geometric factors in cylindrical coordinates.
- Translate continuous operators into conservative numerical forms.

2 Prerequisites and reading strategy

- Basics 2 and multivariable differentiation and integration.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Gradient

For a scalar field f,

$$df = \nabla f \cdot d\mathbf{r}. \tag{3.1}$$

The gradient points toward steepest increase. Pressure force density is $-\nabla p$, so it points toward decreasing pressure. A vector tangent to a level surface $f =$ constant obeys $\mathbf{t} \cdot \nabla f = 0$.

In Cartesian coordinates,

$$\nabla f = \mathbf{e}_x f_x + \mathbf{e}_y f_y + \mathbf{e}_z f_z. \tag{3.2}$$

The directional derivative along unit vector $\mathbf{n}$ is $\mathbf{n} \cdot \nabla f$.

4 Divergence and conservation

Divergence is net outward flux per unit volume:

$$\nabla \cdot \mathbf{F} = \lim_{V \to 0} \frac{1}{V} \oint_{\partial V} \mathbf{F} \cdot d\mathbf{S}. \tag{4.1}$$

The divergence theorem,

$$\int_V \nabla \cdot \mathbf{F} \, dV = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{S}, \tag{4.2}$$

converts an integral balance into a local law. If q is a conserved density with flux $\mathbf{F}$ and source S,

$$\frac{\partial q}{\partial t} + \nabla \cdot \mathbf{F} = S. \tag{4.3}$$

This structure is the mathematical basis of finite-volume methods.

5 Curl and circulation

Stokes theorem states

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\boldsymbol{\ell}. \tag{5.1}$$

Curl is therefore circulation per unit area. Maxwell equations use curl to connect current to magnetic field and changing magnetic flux to electric circulation:

$$\nabla \times \mathbf{B} \approx \mu_0 \mathbf{J}, \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \tag{5.2}$$

Fluid vorticity is $\boldsymbol{\omega} = \nabla \times \mathbf{u}$.

6 Laplacian and diffusion

The scalar Laplacian is

$$\nabla^2 f = \nabla \cdot \nabla f. \tag{6.1}$$

In diffusion,

$$\frac{\partial f}{\partial t} = D \nabla^2 f, \tag{6.2}$$

a local maximum generally has negative Laplacian and decreases, while a local minimum increases. The operator smooths spatial variation. Vector Laplacians require care in curvilinear coordinates because basis vectors also vary.

7 Magnetic pressure and tension identity

A key identity is

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = (\mathbf{B} \cdot \nabla)\mathbf{B} - \nabla \left(\frac{B^2}{2} \right) + \mathbf{B}(\nabla \cdot \mathbf{B}). \tag{7.1}$$

With $\nabla \cdot \mathbf{B} = 0$,

$$\frac{1}{\mu_0}(\nabla \times \mathbf{B}) \times \mathbf{B} = \frac{1}{\mu_0}(\mathbf{B} \cdot \nabla)\mathbf{B} - \nabla \left(\frac{B^2}{2\mu_0} \right). \tag{7.2}$$

The first term is tension along curved field lines; the second is magnetic-pressure force.

8 Induction identity and geometric operators

Another central identity is

$$\nabla \times (\mathbf{u} \times \mathbf{B}) = \mathbf{u}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{u}) \tag{8.1}$$

$$+ (\mathbf{B} \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\mathbf{B}. \tag{8.2}$$

It separates compression, stretching, and advection.

For cylindrical $\mathbf{A} = A_R \mathbf{e}_R + A_\phi \mathbf{e}_\phi + A_Z \mathbf{e}_Z$,

$$\nabla \cdot \mathbf{A} = \frac{1}{R}\frac{\partial}{\partial R} + \frac{1}{R}\frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_Z}{\partial Z}. \tag{8.3}$$

The R factor encodes geometry, not a correction that may be dropped.

A conservative discretization should imitate the integral theorem behind the differential operator. Discrete divergence should be net face flux divided by cell volume.

9 Computational laboratory

On a Cartesian grid, define an analytic vector field and compute its divergence and curl exactly and numerically. Demonstrate second-order convergence. Verify the divergence theorem by comparing volume integral and boundary flux. Repeat for an axisymmetric cylindrical field and show the error from omitting the $1/R$ factor.

10 Summary

- Gradient measures directional scalar change.

- Divergence is the local operator of conservation laws.
- Curl measures circulation and connects currents and induction.
- Vector identities reveal magnetic pressure, tension, stretching, and advection.
- Geometry must be built into both analytic and discrete operators.

Symbols and acronyms used in this document

Symbol / acronym	**Name**	**Meaning in this document**
∇f	gradient	Steepest scalar increase.
$\nabla \cdot \mathbf{F}$	divergence	Net outward flux per volume.
$\nabla \times \mathbf{F}$	curl	Local circulation density.
∇^2	Laplacian	Divergence of gradient.
$d\mathbf{S}$	oriented area	Normal direction times surface area.

11 Exercises

1. Use the divergence theorem with $\mathbf{F} = \mathbf{r}/3$ to derive the volume of a sphere.
2. Prove $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for smooth Cartesian fields.
3. Derive the magnetic pressure-tension identity using index notation.
4. Compute divergence and curl of rigid rotation $\mathbf{u} = (-y, x, 0)$.
5. Write the integral form of the continuity equation for a control volume.

12 References

References

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